

The empirical recurrence for OEIS A259296: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

2 September 2026

Abstract

OEIS A259296 records “Empirical recurrence of order 72”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 73$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 72, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 72 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A259296 is “Number of $(n+1) \times (7+1)$ 0..1 arrays with each 2×2 subblock having clockwise pattern 0001 0101 or 0111.”. It has offset 1 and begins

2850, 48002, 688966, 10862998, 165147350, 2569298018, 39630793526, 614892766122, ...

The entry states:

Empirical recurrence of order 72 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 11:50:35, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 257$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 73$ of the 257, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 73$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 146$ exact terms of the model returns order 72 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{72} c_i a(n-i),$$

c_1	10	c_{19}	−412402751422155936	c_{37}	81721395507593550942849910	c_{55}	−
c_2	494	c_{20}	−763035235687276342	c_{38}	21187257726977847103660344	c_{56}	−
c_3	−4474	c_{21}	7137765806391744722	c_{39}	−287833586363826421446096112	c_{57}	−
c_4	−104956	c_{22}	10470681398042559236	c_{40}	−60696365948716538196606496	c_{58}	−
c_5	897722	c_{23}	−101600037706826598138	c_{41}	854267468530399946073842592	c_{59}	−
c_6	12962448	c_{24}	−11889818522230983904	c_{42}	148026419267678164165423488	c_{60}	−
c_7	−107663432	c_{25}	1199061429195296313736	c_{43}	−2131253150669695484690694656	c_{61}	−
c_8	−1056352662	c_{26}	1124697658525661982110	c_{44}	−309284256715865097534638848	c_{62}	−
c_9	8684715568	c_{27}	−11806785171778264836992	c_{45}	4454358808541369827540578816	c_{63}	−
c_{10}	60993775236	c_{28}	−8906693442851927724612	c_{46}	558269964290559244052877312	c_{64}	−
c_{11}	−503173270822	c_{29}	97459998008975175642166	c_{47}	−7764294728819147735198904320	c_{65}	−
c_{12}	−2613060028874	c_{30}	59261366707575273455986	c_{48}	−877592911299942331436851200	c_{66}	−
c_{13}	21847403682580	c_{31}	−676736385103805132158956	c_{49}	11223194089787936970354909184	c_{67}	−
c_{14}	85805306234546	c_{32}	−332102109626380334294834	c_{50}	1205322897864089051051589632	c_{68}	−
c_{15}	−732525952499112	c_{33}	3961930497365763820150360	c_{51}	−13357892275617021024939671552	c_{69}	−
c_{16}	−2212276272339740	c_{34}	1570149410290604960625564	c_{52}	−1437327786516242028428787712	c_{70}	−
c_{17}	19393751830680394	c_{35}	−19581020979489553398537314	c_{53}	12975317055985710053323702272	c_{71}	−
c_{18}	45615737664589662	c_{36}	−6270968491100877791237357	c_{54}	1462927390290788393297117184	c_{72}	−

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 72, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $72 + 4$ terms removed returns order 72 as well, so $\deg q_{\min} = 72 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 14 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $72 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 72 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A259296.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.